

复合函数求导法则

先回忆一下一元复合函数的微分法则

若 $y = f(u)$ 而 $u = \varphi(x)$ 可导 则复合函数

$y = f[\varphi(x)]$ 对 x 的导数为 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

这一节我们将把这一求导法则推广到多元函数的情形，主要介绍多元复合函数的微分法和隐函数的微分法。我们知道，求偏导数与求一元函数的导数本质上并没有区别，对一元函数适用的微分法包括复合函数的微分法在内，在多元函数微分法中仍然适用，那么为什么还要介绍多元

复合函数的微分法和隐函数的微分法呢？

这主要是对于没有具体给出式子的所谓抽象函数

如 $z = f(x^2 - y^2, xy)$ 它是由 $z = f(u, v)$

及 $u = x^2 - y^2, v = xy$ 复合而成的

由于 f 没有具体给出 在求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ 时

一元复合函数的微分法则就无能为力了，为此还要介绍多元复合函数的微分法和隐函数的微分法。

一、链式法则

定理 如果函数 $u = \phi(t)$ 及 $v = \psi(t)$ 都在点 t 可导, 函数 $z = f(u, v)$ 在对应点 (u, v) 具有连续偏导数, 则复合函数 $z = f[\phi(t), \psi(t)]$ 在对应点 t 可导, 且其导数可用下列公式计算:

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt}.$$

证 设 t 获得增量 Δt ,

则 $\Delta u = \phi(t + \Delta t) - \phi(t)$, $\Delta v = \psi(t + \Delta t) - \psi(t)$;

由于函数 $z = f(u, v)$ 在点 (u, v) 有连续偏导数

$$\Delta z = \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + \varepsilon_1 \Delta u + \varepsilon_2 \Delta v,$$

当 $\Delta u \rightarrow 0$, $\Delta v \rightarrow 0$ 时, $\varepsilon_1 \rightarrow 0$, $\varepsilon_2 \rightarrow 0$

$$\frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial u} \cdot \frac{\Delta u}{\Delta t} + \frac{\partial z}{\partial v} \cdot \frac{\Delta v}{\Delta t} + \varepsilon_1 \frac{\Delta u}{\Delta t} + \varepsilon_2 \frac{\Delta v}{\Delta t}$$

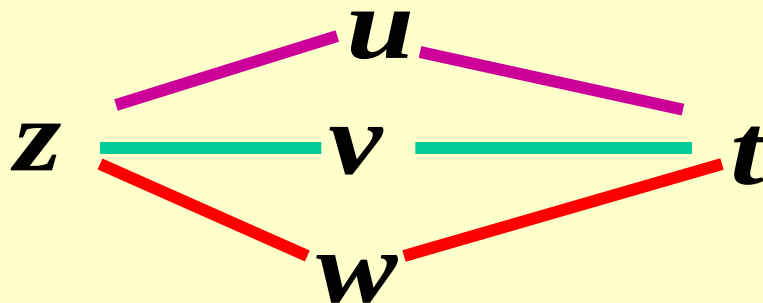
当 $\Delta t \rightarrow 0$ 时, $\Delta u \rightarrow 0$, $\Delta v \rightarrow 0$

$$\frac{\Delta u}{\Delta t} \rightarrow \frac{du}{dt}, \quad \frac{\Delta v}{\Delta t} \rightarrow \frac{dv}{dt},$$

$$\frac{dz}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}.$$

上定理的结论可推广到中间变量多于两个的情况.

如
$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt} + \frac{\partial z}{\partial w} \frac{dw}{dt}$$



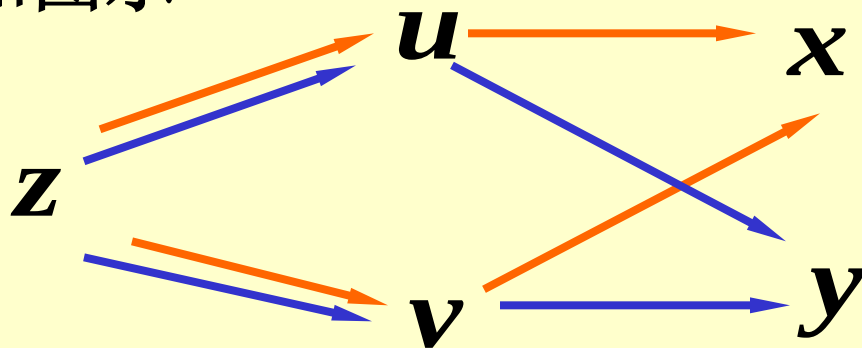
以上公式中的导数 $\frac{dz}{dt}$ 称为**全导数**.

上定理还可推广到中间变量不是一元函数而是多元函数的情况: $z = f[\phi(x, y), \psi(x, y)]$.

如果 $u = \phi(x, y)$ 及 $v = \psi(x, y)$ 都在点 (x, y) 具有对 x 和 y 的偏导数, 且函数 $z = f(u, v)$ 在对应点 (u, v) 具有连续偏导数, 则复合函数 $z = f[\phi(x, y), \psi(x, y)]$ 在对应点 (x, y) 的两个偏导数存在, 且可用下列公式计算

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y}.$$

链式法则如图示



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}.$$

称为标准法则或 2×2 法则

这个公式的特征:

(1) 函数 $z = f[u(x, y), v(x, y)]$ 有两个自变量 x 和 y

故法则中包含 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ 两个公式;

(2)由于在复合过程中有两个中间变量 u 和 v

故法则中每一个公式都是两项之和，这两项分别含有 $\frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}$

(3)每一项的构成与一元复合函数的链导法则类似，

即“函数对中间变量的导数乘以中间变量对自变量的导数”

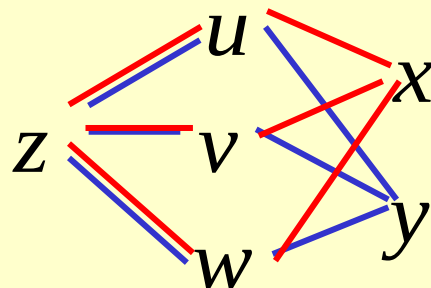
多元复合函数的求导法则简言之即：

“分道相加，连线相乘”

类似地再推广，设 $u = \phi(x, y)$ 、 $v = \psi(x, y)$ 、 $w = w(x, y)$ 都在点 (x, y) 具有对 x 和 y 的偏导数，复函数 $z = f[\phi(x, y), \psi(x, y), w(x, y)]$ 在对应点 (x, y) 两个偏导数存在，且可用下列公式计算

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial z}{\partial w} \frac{\partial w}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial z}{\partial w} \frac{\partial w}{\partial y}.$$



特殊地 $z = f(u, x, y)$ 其中 $u = \phi(x, y)$

即 $z = f[\phi(x, y), x, y]$, 令 $v = x$, $w = y$,

$$\frac{\partial v}{\partial x} = 1, \quad \frac{\partial w}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 0, \quad \frac{\partial w}{\partial y} = 1.$$

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial y}.$$

区别类似

两者的区别

把复合函数 $z = f[\phi(x, y), x, y]$ 中的 y 看作不变而对 x 的偏导数

把 $z = f(u, x, y)$ 中的 u 及 y 看作不变而对 x 的偏导数

注 此公式可以推广到任意多个中间变量和任意多个自变量的情形

$$\text{如 } z = f(u_1, u_2, \cdots, u_m) \quad u_i = u_i(x_1, x_2, \cdots, x_n) \\ (i = 1, 2, \cdots, m)$$

则

$$\frac{\partial z}{\partial x_j} = \sum_{i=1}^m \frac{\partial z}{\partial u_i} \cdot \frac{\partial u_i}{\partial x_j}, (j = 1, 2, \cdots, n)$$

从以上推广中我们可以得出：所有公式中两两乘积的项数等于中间变量的个数，而与自变量的个数无关

关于多元复合函数求偏导问题

这是一项基本技能，要求熟练掌握，尤其是求二阶偏导数，既是重点又是难点。对求导公式不求强记，而要切实做到彻底理解。注意以下几点将会有助于领会和理解公式，在解题时自如地运用公式

①用图示法表示出函数的复合关系

②函数对某个自变量的偏导数的结构

（项数及项的构成）

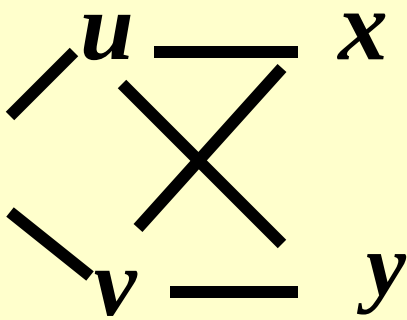
③弄清 $f_u(u,v), f_v(u,v)$ 的结构是求抽象的复合函数的二阶偏导数的关键

$f_u(u,v), f_v(u,v)$ 仍是复合函数

且复合结构与原来的 $f(u, v)$ 完全相同

即仍是以 u, v 为中间变量, 以 x, y 为自变量的复合函数

因此求它们关于 x, y 的偏导数时必须使链式法则

$$\frac{\partial z}{\partial u} = f_u(u, v)$$


$$\frac{\partial}{\partial x} [f_u(u, v)] = f_{uu} \frac{\partial u}{\partial x} + f_{uv} \frac{\partial v}{\partial x}$$

$$\frac{\partial}{\partial x} [f_v(u, v)] = f_{vu} \frac{\partial u}{\partial x} + f_{vv} \frac{\partial v}{\partial x}$$

在具体计算中最容易出错的地方是对

$f_u(u,v)$ 再求偏导数这一步

原因就是不注意 $f''(u,v)$ 是与 $f(u,v)$ 具有相同结构的复合函数易被误认为仅是 u 的函数，从而导致漏掉 f_{uv} 这一项

④求抽象函数的偏导数时，一定要设中间变量

⑤注意引用这些公式的条件

外层函数可微（偏导数连续）内层函数可导

⑥ f''_{uv}, f''_{vu} 的合并问题 视题设条件

例 1 设 $z = e^u \sin v$, 而 $u = xy$, $v = x + y$,

求 $\frac{\partial z}{\partial x}$ 和 $\frac{\partial z}{\partial y}$.

解

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= e^u \sin v \cdot y + e^u \cos v \cdot 1 = e^u (y \sin v + \cos v),$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= e^u \sin v \cdot x + e^u \cos v \cdot 1 = e^u (x \sin v + \cos v).$$

例 2 设 $z = uv + \sin t$, 而 $u = e^t$, $v = \cos t$,

求全导数 $\frac{dz}{dt}$.

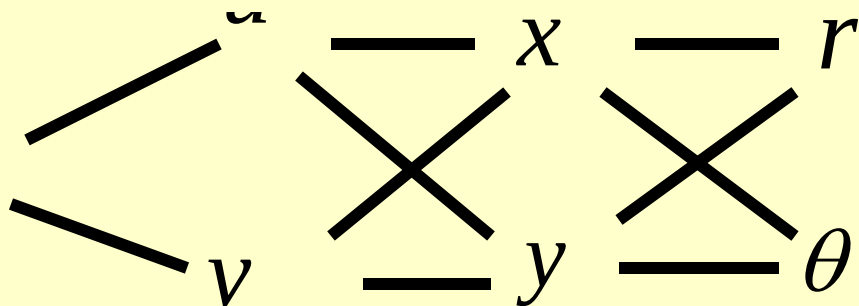
解

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial t} \\ &= ve^t - u \sin t + \cos t = e^t \cos t - e^t \sin t + \cos t \\ &= e^t (\cos t - \sin t) + \cos t.\end{aligned}$$

例3 设 $w = f(u, v), u = u(x, y), v = v(x, y),$
 $x = x(r, \theta), y = y(r, \theta)$

均满足复合函数求偏导数的条件 计算 $\frac{\partial w}{\partial r}, \frac{\partial w}{\partial \theta}$
 (两重复合问题)

解 由链式法则



$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial u} \cdot \frac{\partial u}{\partial r} + \frac{\partial w}{\partial v} \cdot \frac{\partial v}{\partial r} \quad \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial r}$$

故

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial u} \left(\frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} \right) + \frac{\partial w}{\partial v} \left(\frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial r} \right)$$

同理可得

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial u} \left(\frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta} \right) + \frac{\partial w}{\partial v} \left(\frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial \theta} \right)$$

例 4 设 $w = f(x + y + z, xyz)$, f 具有二阶

连续偏导数, 求 $\frac{\partial w}{\partial x}$ 和 $\frac{\partial^2 w}{\partial x \partial z}$.

解 令 $u = x + y + z$, $v = xyz$;

$$\text{记 } f'_1 = \frac{\partial f(u, v)}{\partial u}, \quad f''_{12} = \frac{\partial^2 f(u, v)}{\partial u \partial v},$$

同理有 f'_2 , f''_{11} , f''_{22} .

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} = f'_1 + yzf'_2;$$

$$\frac{\partial^2 w}{\partial x \partial z} = \frac{\partial}{\partial z} (f'_1 + yzf'_2) = \frac{\partial f'_1}{\partial z} + yf'_2 + yz \frac{\partial f'_2}{\partial z};$$

$$\frac{\partial f'_1}{\partial z} = \frac{\partial f'_1}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f'_1}{\partial v} \cdot \frac{\partial v}{\partial z} = f''_{11} + xyf''_{12};$$

$$\frac{\partial f'_2}{\partial z} = \frac{\partial f'_2}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f'_2}{\partial v} \cdot \frac{\partial v}{\partial z} = f''_{21} + xyf''_{22};$$

$$\begin{aligned} \text{于是 } \frac{\partial^2 w}{\partial x \partial z} &= f''_{11} + xyf''_{12} + yf'_2 + yz(f''_{21} + xyf''_{22}) \\ &= f''_{11} + y(x+z)f''_{12} + xy^2zf''_{22} + yf'_2. \end{aligned}$$

二、全微分形式不变性

设函数 $z = f(u, v)$ 具有连续偏导数, 则有全微分

$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv; \text{当 } u = \phi(x, y)、v = \psi(x, y)$$

时, 有 $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$

全微分形式不变形的实质:

无论 z 是自变量 u 、 v 的函数或中间变量 u 、 v 的函数, 它的全微分形式是一样的.

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$= \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) dx + \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) dy$$

$$\begin{aligned} &= \frac{\partial z}{\partial u} \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + \frac{\partial z}{\partial v} \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right) \\ &= \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv. \end{aligned}$$

利用全微分形式不变性，在逐步作微分运算的过程中，不论变量间的关系如何错综复杂，都可以不加辨认和区分，而一律作为自变量来处理

且作微分运算的结果对自变量的微分

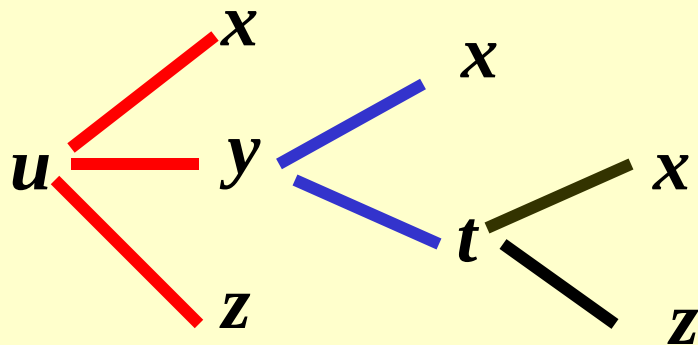
$dx, dy, dz, \dots$ 来说是线性的

从而为解题带来很多方便，而且也不易出错

例5 设 $u = f(x, y, z), y = \varphi(x, t), t = \psi(x, z)$

各函数满足求导条件 求 $\frac{\partial u}{\partial x}$

解一 变量间的关系如下图所示



$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} \quad \frac{\partial y}{\partial x} = \frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial t} \cdot \frac{\partial \psi}{\partial x}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{\partial \varphi}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{\partial \varphi}{\partial t} \cdot \frac{\partial \psi}{\partial x}$$

解二 这里变量间的关系比较混乱

用全微分来解 由全微分定理

$$\begin{aligned} du &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \\ &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} \left[\frac{\partial \varphi}{\partial x} dx + \frac{\partial \varphi}{\partial t} dt \right] + \frac{\partial f}{\partial z} dz \\ &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} \left[\frac{\partial \varphi}{\partial x} dx + \frac{\partial \varphi}{\partial t} \left(\frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial z} dz \right) \right] + \frac{\partial f}{\partial z} dz \end{aligned}$$

注意到 x, z 是独立自变量

故

$$du = \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial t} \frac{\partial \psi}{\partial x} \right) dx + \left(\frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial t} \frac{\partial \psi}{\partial z} + \frac{\partial f}{\partial z} \right) dz$$

由全微分定义

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial t} \frac{\partial \psi}{\partial x}$$

$$\frac{\partial u}{\partial z} = \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial t} \frac{\partial \psi}{\partial z} + \frac{\partial f}{\partial z}$$

注 解法二在实际计算中显得十分灵便且不易出错

三、小结

1、链式法则（分三种情况）

（特别要注意课中所讲的特殊情况）

2、全微分形式不变性

（理解其实质）

思考题

设 $z = f(u, v, x)$, 而 $u = \phi(x)$, $v = \psi(x)$,

$$\text{则 } \frac{dz}{dx} = \frac{\partial f}{\partial u} \frac{du}{dx} + \frac{\partial f}{\partial v} \frac{dv}{dx} + \frac{\partial f}{\partial x},$$

试问 $\frac{dz}{dx}$ 与 $\frac{\partial f}{\partial x}$ 是否相同? 为什么?

思考题解答

不相同.

等式左端的 z 是作为一个自变量 x 的函数,

而等式右端最后一项 f 是作为 u, v, x 的三元函数,

写出来为

$$\left. \frac{dz}{dx} \right|_x = \left. \frac{\partial f}{\partial u} \right|_{(u,v,x)} \cdot \left. \frac{du}{dx} \right|_x + \left. \frac{\partial f}{\partial v} \right|_{(u,v,x)} \cdot \left. \frac{dv}{dx} \right|_x + \left. \frac{\partial f}{\partial x} \right|_{(u,v,x)}.$$

练习题

一、填空题：

1、设 $z = \frac{x \cos y}{y \cos x}$, 则 $\frac{\partial z}{\partial x} =$ _____;

$$\frac{\partial z}{\partial y} = \text{_____}.$$

2、设 $z = \frac{x^2 \ln(3x - 2y)}{y^2}$, 则 $\frac{\partial z}{\partial x} =$ _____;

$$\frac{\partial z}{\partial y} = \text{_____}.$$

3、设 $z = e^{\sin t - 2t^3}$, 则 $\frac{dz}{dt} =$ _____.

二、设 $z = ue^{\frac{v}{u}}$, 而 $u = x^2 + y^2, v = xy$, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

三、设 $z = \arctan(xy)$, 而 $y = e^x$, 求 $\frac{dz}{dx}$.

四、设 $z = f(x^2 - y^2, e^{xy})$, (其中 f 具有一阶连续偏导数), 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

五、设 $u = f(x + xy + xyz)$, (其中 f 具有一阶连续偏导数), 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$.

六、设 $z = f(x, \frac{x}{y})$, (其中 f 具有二阶连续偏导数), 求 $\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2}$.

七、设 $z = \frac{y}{f(x^2 - y^2)}$, 其中 f 为可导函数,

验证: $\frac{1}{x} \frac{\partial z}{\partial x} + \frac{1}{y} \frac{\partial z}{\partial y} = \frac{z}{y^2}.$

八、设 $z = \phi[x + \varphi(x - y), y]$, 其中 ϕ, φ 具有二阶导数, 求

$$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}.$$

练习题答案

一、 1、 $\frac{\cos y(\cos x + x \sin x)}{y \cos^2 x}, -\frac{x \cos x(y \sin y + \cos y)}{y^2 \cos^2 x};$

2、 $\frac{2x}{y^2} \ln(3x - 2y) + \frac{3x^2}{(3x - 2y)y^2},$
 $-\frac{2x^2}{y^3} \ln(3x - 2y) - \frac{2x^2}{(3x - 2y)y^2};$

3、 $\frac{3(1 - 4t^2)}{\sqrt{1 - (3t - 4t^3)^2}}.$

二、 $\frac{\partial z}{\partial x} = [2x + y - \frac{2x^2 y}{(x^2 + y^2)y^2}]e^{\frac{xy}{x^2 + y^2}},$

$$\frac{\partial z}{\partial y} = [2y + x - \frac{2y^2 x}{(x^2 + y^2)}]e^{\frac{xy}{(x^2 + y^2)}}.$$

$$\text{三、} \frac{dz}{dx} = \frac{e^x(1+x)}{1+x^2e^{2x}}.$$

$$\text{四、} \frac{\partial z}{\partial x} = 2xf_1' + ye^{xy}f_2', \frac{\partial z}{\partial y} = -2yf_1' + xe^{xy}f_2'.$$

$$\text{五、} \frac{\partial u}{\partial x} = f'(1+y+yz), \frac{\partial u}{\partial y} = f'(x+xz), \frac{\partial u}{\partial z} = xyf'.$$

$$\text{六、} \frac{\partial^2 z}{\partial x^2} = f_{11}'' + \frac{2}{y} f_{12}'' + \frac{1}{y^2} f_{22}'',$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{x}{y^2} (f_{12}'' + \frac{1}{y} f_{22}'') - \frac{1}{y^2} f_2',$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{2x}{y^3} f_2' + \frac{x^2}{y^4} f_{22}''.$$

$$\text{八、 } \frac{\partial^2 z}{\partial x^2} = \phi_{11}(1 + \phi')^2 + \phi_1 \phi'',$$

$$\frac{\partial^2 z}{\partial y^2} = \phi_{11}(\phi')^2 - \phi_{12}\phi' + \phi_1\phi'' - \phi_{21}\phi' + \phi_{22}.$$